
SmartAuto Post

AI Powered Social Media Automation Platform

Part 1: Backend Master Blueprint deta hoon. Isko aap project ka foundation document samjho.

Version

v1.0 MVP Architecture

Prepared For

Development Team
Backend Engineers
AI Engineers
DevOps Engineers

1. PROJECT OVERVIEW

Product Name

SmartAuto Post

Product Category

AI Powered Social Media Automation SaaS

Goal

Allow businesses, agencies, and creators to:

Generate Content
Generate Hashtags
Generate SEO Text
Generate Images
Create Content Calendar
Schedule Posts
Publish Automatically
Track Analytics
Manage Multiple Clients

Without opening:

Facebook
Instagram
LinkedIn
Threads
Google Business

2. CORE BUSINESS FLOW

User Register
↓
Create Organization
↓
Connect Social Accounts
↓
Setup Brand Profile
↓
Enter Keywords
↓
Generate Content
↓
Generate Hashtags
↓
Generate Images
↓
Generate Calendar
↓
Approve Schedule
↓
Auto Publish
↓
Collect Analytics
↓
AI Optimization Suggestions

3. TECH STACK

Frontend

Next.js
TypeScript
TailwindCSS
ShadCN UI

Backend

FastAPI

Python 3.13+

Database

PostgreSQL

Cache

Redis

Queue

Celery

Scheduler

Celery Beat

Storage

AWS S3

or

Cloudflare R2

Authentication

JWT

Refresh Token

AI Engine

OpenAI

Llama Fallback

4. SYSTEM MODULES

Main Services:

Auth Service

Organization Service

User Service

Social Service

Content AI Service

Media Service

Scheduler Service
Analytics Service
Billing Service
Notification Service
Admin Service

5. AUTH MODULE

Purpose:

User authentication
Authorization
Session management

Features:

Register
Login
Logout
Forgot Password
Reset Password
Email Verification
2FA
Social Login

Database:

users

id
name
email
password_hash
status
is_verified
created_at
updated_at

refresh_tokens

id
user_id
token
expires_at

6. ORGANIZATION MODULE

One company = one organization

Example:

Teknowxa
ABC Digital
XYZ Media

Table:

organizations

id
name
slug
logo
industry
timezone
language
owner_id

7. TEAM MODULE

Roles:

Owner
Admin
Manager
Editor
Viewer

Table:

organization_users

id
organization_id
user_id
role
status

8. BRAND KIT MODULE

Stores branding assets

Table:

brand_kits

id
organization_id
logo_url
primary_color
secondary_color
font_family
watermark_enabled

9. SOCIAL ACCOUNT MODULE

Supported Platforms

Facebook
Instagram
LinkedIn
Threads
Google Business

Table:

social_accounts

id
organization_id
platform
account_name
account_id
access_token
refresh_token
token_expiry
status

Requirements:

OAuth Flow
Token Refresh
Permission Validation

10. KEYWORD ENGINE

Input:

CRM Software
ERP Software
School Admission

Output:

Topic Clusters
Related Keywords
Content Ideas

Table:

keywords

id
organization_id
keyword
industry
language

11. CONTENT AI ENGINE

Generate:

Captions
Hashtags
SEO Text
CTA
Carousel Ideas
Video Ideas
Post Ideas

Table:

generated_contents

id
organization_id
keyword_id
type
content
language

Types:

caption
hashtag
seo
cta
calendar

12. IMAGE ENGINE

Support:

Upload
Template
AI Generated
Stock Images

Table:

media_assets

id
organization_id
type
url
thumbnail
created_at

13. POST MODULE

Central posting system

Table:

posts

id
organization_id
title
caption
image_url
status
created_by

Status:

draft
scheduled
published
failed

14. POST PLATFORM TABLE

One post can publish to multiple platforms

Table:

post_platforms

```
id
post_id
platform
platform_post_id
status
response
```

15. CALENDAR MODULE

Stores schedules

Table:

post_schedules

```
id
post_id
schedule_time
timezone
status
```

16. PUBLISHING ENGINE

Flow:

```
Scheduler detects post
↓
Create Queue Job
↓
Select Provider
↓
Call Platform API
↓
Save Response
↓
Update Status
```

Providers:

```
FacebookProvider
InstagramProvider
```

LinkedInProvider
ThreadsProvider
GoogleBusinessProvider

17. ANALYTICS MODULE

Collect:

Reach
Impressions
Likes
Comments
Shares
Clicks
Followers

Tables:

analytics_logs

id
post_id
platform
metric_type
metric_value
recorded_at

18. COMPETITOR MODULE

Track:

Posting Frequency
Top Hashtags
Top Posts
Engagement

Tables:

competitors

competitor_metrics

19. FESTIVAL AUTOMATION MODULE

Tables:

festival_master

id
festival_name
festival_date
country

festival_templates

id
industry
caption
image_template
hashtags

20. NOTIFICATION MODULE

Channels:

Email
In App
WhatsApp (future)

Events:

Post Published
Post Failed
Subscription Expiring
New Analytics Report

21. BILLING MODULE

Plans:

Free
Pro
Agency
Enterprise

Tables:

plans

subscriptions

payments

usage_logs

22. ADMIN PANEL

Admin Controls:

Users
Organizations
Subscriptions
Payments
Feature Flags
AI Usage
API Usage
Analytics
Revenue

23. API STRUCTURE

Main API Groups:

/api/auth/*
/api/users/*
/api/organizations/*
/api/social/*
/api/content/*
/api/posts/*
/api/scheduler/*
/api/analytics/*
/api/subscriptions/*
/api/admin/*

24. QUEUE JOBS

Celery Workers:

GenerateCaptionJob
GenerateHashtagJob
GenerateSEOJob
GenerateCalendarJob
PublishPostJob
RefreshTokenJob
FetchAnalyticsJob

25. SECURITY REQUIREMENTS

Mandatory:

JWT Authentication
RBAC
Rate Limiting
Encrypted Tokens
HTTPS Only
Audit Logs
2FA Support

26. MVP RELEASE SCOPE

Version 1:

Auth
Organizations
Brand Kit
Facebook Connect
LinkedIn Connect
Caption Generator
Hashtag Generator
Post Scheduler
Publishing Engine
Analytics Dashboard
Subscription System
Admin Panel

SmartAuto Post

Part-2: Database Architecture & ER Design Document

Version: 1.0

Purpose: Production-grade PostgreSQL database design for AI-powered social media automation SaaS.

DATABASE DESIGN PRINCIPLES

System must support:

Multi-Tenant SaaS
Organizations
Teams
Clients
Multi Platform Posting
AI Content Generation
Scheduler
Analytics
Billing
Admin Panel

Database Type:

PostgreSQL 16+

Naming Convention:

snake_case

Primary Keys:

uuid

MODULE 1: USER MANAGEMENT

users

id UUID PK
name VARCHAR(255)
email VARCHAR(255) UNIQUE
password_hash TEXT
phone VARCHAR(20)
avatar_url TEXT

email_verified BOOLEAN
two_factor_enabled BOOLEAN

status VARCHAR(50)

created_at TIMESTAMP
updated_at TIMESTAMP

user_sessions

```
id UUID PK
user_id UUID FK

device_name VARCHAR(255)
ip_address VARCHAR(255)

access_token TEXT
refresh_token TEXT

expires_at TIMESTAMP

created_at TIMESTAMP
```

MODULE 2: ORGANIZATION

organizations

```
id UUID PK

name VARCHAR(255)
slug VARCHAR(255)

industry VARCHAR(255)

logo_url TEXT

timezone VARCHAR(255)

language VARCHAR(50)

owner_id UUID

created_at TIMESTAMP
updated_at TIMESTAMP
```

organization_users

```
id UUID PK

organization_id UUID
user_id UUID

role VARCHAR(50)

status VARCHAR(50)

created_at TIMESTAMP
```

Roles:

owner
admin
manager
editor
viewer

MODULE 3: CLIENT MANAGEMENT

clients

id UUID PK
organization_id UUID
client_name VARCHAR(255)
industry VARCHAR(255)
logo_url TEXT
status VARCHAR(50)
created_at TIMESTAMP

client_users

id UUID PK
client_id UUID
user_id UUID
role VARCHAR(50)
created_at TIMESTAMP

MODULE 4: BRAND KIT

brand_kits

id UUID PK
organization_id UUID

```
logo_url TEXT

primary_color VARCHAR(20)

secondary_color VARCHAR(20)

font_family VARCHAR(255)

watermark_enabled BOOLEAN

created_at TIMESTAMP
```

MODULE 5: SOCIAL ACCOUNTS

social_accounts

```
id UUID PK

organization_id UUID

client_id UUID NULL

platform VARCHAR(100)

account_name VARCHAR(255)

account_id VARCHAR(255)

access_token TEXT

refresh_token TEXT

token_expiry TIMESTAMP

status VARCHAR(50)

created_at TIMESTAMP
```

Platforms:

```
facebook
instagram
linkedin
threads
google_business
```

MODULE 6: KEYWORDS

keywords

```
id UUID PK
organization_id UUID
keyword TEXT
industry VARCHAR(255)
language VARCHAR(50)
created_by UUID
created_at TIMESTAMP
```

MODULE 7: CONTENT GENERATION

generated_contents

```
id UUID PK
organization_id UUID
keyword_id UUID
content_type VARCHAR(100)
title TEXT
content LONGTEXT
language VARCHAR(50)
ai_model VARCHAR(100)
created_by UUID
created_at TIMESTAMP
```

Types:

```
caption
seo
hashtags
cta
post_idea
calendar
```

hashtags

id UUID PK

content_id UUID

hashtag VARCHAR(255)

category VARCHAR(100)

created_at TIMESTAMP

Categories:

primary

industry

trending

MODULE 8: MEDIA LIBRARY

media_assets

id UUID PK

organization_id UUID

client_id UUID

file_name VARCHAR(255)

file_type VARCHAR(50)

file_url TEXT

thumbnail_url TEXT

size BIGINT

created_by UUID

created_at TIMESTAMP

image_templates

id UUID PK

template_name VARCHAR(255)

```
industry VARCHAR(255)

preview_url TEXT

template_json JSONB

created_at TIMESTAMP
```

MODULE 9: POSTS

posts

```
id UUID PK

organization_id UUID

client_id UUID

title TEXT

caption LONGTEXT

image_url TEXT

status VARCHAR(50)

created_by UUID

created_at TIMESTAMP
updated_at TIMESTAMP
```

Status:

```
draft
scheduled
published
failed
archived
```

post_platforms

```
id UUID PK

post_id UUID

platform VARCHAR(100)

platform_post_id VARCHAR(255)
```

```
status VARCHAR(50)

response JSONB

published_at TIMESTAMP
```

MODULE 10: SCHEDULER

post_schedules

```
id UUID PK

post_id UUID

schedule_time TIMESTAMP

timezone VARCHAR(255)

status VARCHAR(50)

retry_count INTEGER

created_at TIMESTAMP
```

Status:

```
pending
processing
published
failed
```

scheduler_logs

```
id UUID PK

post_schedule_id UUID

message TEXT

log_type VARCHAR(50)

created_at TIMESTAMP
```

MODULE 11: ANALYTICS

post_metrics

```
id UUID PK
post_id UUID
platform VARCHAR(100)
likes INTEGER
comments INTEGER
shares INTEGER
clicks INTEGER
reach INTEGER
impressions INTEGER
recorded_at TIMESTAMP
```

follower_metrics

```
id UUID PK
social_account_id UUID
followers_count INTEGER
recorded_at TIMESTAMP
```

analytics_summary

```
id UUID PK
organization_id UUID
period VARCHAR(50)
total_posts INTEGER
total_reach BIGINT
total_clicks BIGINT
engagement_rate DECIMAL
generated_at TIMESTAMP
```

MODULE 12: COMPETITOR TRACKING

competitors

```
id UUID PK
organization_id UUID
platform VARCHAR(100)
profile_name VARCHAR(255)
profile_url TEXT
created_at TIMESTAMP
```

competitor_metrics

```
id UUID PK
competitor_id UUID
followers INTEGER
engagement_rate DECIMAL
posts_count INTEGER
recorded_at TIMESTAMP
```

MODULE 13: FESTIVAL AUTOMATION

festival_master

```
id UUID PK
festival_name VARCHAR(255)
festival_date DATE
country VARCHAR(100)
created_at TIMESTAMP
```

festival_templates

id UUID PK
festival_id UUID
industry VARCHAR(255)
caption TEXT
hashtags TEXT
image_template TEXT
created_at TIMESTAMP

MODULE 14: NOTIFICATIONS

notifications

id UUID PK
user_id UUID
title VARCHAR(255)
message TEXT
channel VARCHAR(50)
is_read BOOLEAN
created_at TIMESTAMP

Channels:

email
push
in_app
whatsapp

MODULE 15: BILLING

plans

id UUID PK
plan_name VARCHAR(255)
price DECIMAL

```
max_accounts INTEGER  
max_posts INTEGER  
max_clients INTEGER  
max_team_members INTEGER  
created_at TIMESTAMP
```

subscriptions

```
id UUID PK  
organization_id UUID  
plan_id UUID  
start_date DATE  
end_date DATE  
status VARCHAR(50)  
created_at TIMESTAMP
```

payments

```
id UUID PK  
subscription_id UUID  
amount DECIMAL  
payment_gateway VARCHAR(100)  
transaction_id VARCHAR(255)  
status VARCHAR(50)  
created_at TIMESTAMP
```

MODULE 16: AI USAGE TRACKING

ai_requests

```
id UUID PK
organization_id UUID
request_type VARCHAR(100)
prompt_tokens INTEGER
completion_tokens INTEGER
cost DECIMAL
response_time_ms INTEGER
created_at TIMESTAMP
```

Request Types:

```
caption
hashtags
seo
calendar
image_prompt
competitor_analysis
```

MODULE 17: SYSTEM AUDIT LOGS

audit_logs

```
id UUID PK
user_id UUID
action VARCHAR(255)
module VARCHAR(255)
old_value JSONB
new_value JSONB
ip_address VARCHAR(255)
created_at TIMESTAMP
```

MODULE 18: ADMIN PANEL

admin_users

```
id UUID PK
name VARCHAR(255)
email VARCHAR(255)
password_hash TEXT
role VARCHAR(50)
created_at TIMESTAMP
```

feature_flags

```
id UUID PK
feature_name VARCHAR(255)
enabled BOOLEAN
created_at TIMESTAMP
```

Examples:

```
ai_image_generation
competitor_tracking
whatsapp_broadcast
festival_automation
```

CRITICAL INDEXES

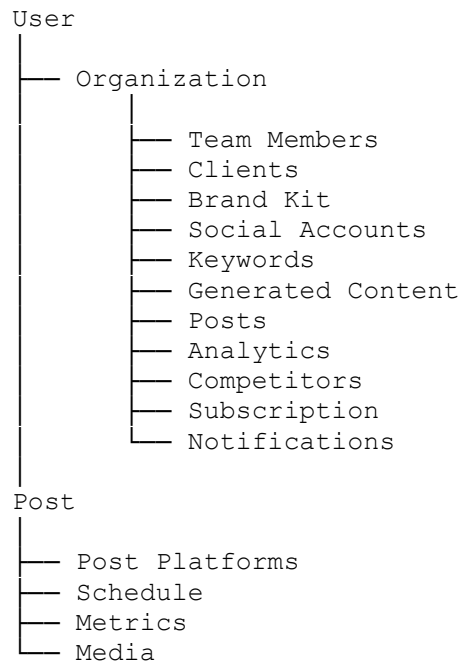
Must create indexes on:

```
users(email)
organizations(slug)
social_accounts(platform)
posts(status)
posts(created_at)
post_schedules(schedule_time)
post_schedules(status)
post_platforms(platform)
analytics_summary(organization_id)
```

subscriptions(organization_id)

payments(transaction_id)

HIGH LEVEL ER RELATIONSHIP



SmartAuto Post

Part-3: API Specification Document (OpenAPI / Swagger Level)

Version: 1.0

Purpose: Complete API contract for Frontend, Mobile App, Admin Panel, AI Engine, Scheduler, and Third-Party Integrations.

API ARCHITECTURE

Base URL

`https://api.smartautopost.com/api/v1`

Authentication

Authorization: Bearer JWT_TOKEN

Response Format

Success:

```
{
  "success": true,
  "message": "Request successful",
  "data": {}
}
```

Error:

```
{
  "success": false,
  "message": "Validation failed",
  "errors": []
}
```

MODULE 1: AUTH APIs

Register User

POST /auth/register

Request

```
{
  "name": "Rakesh",
  "email": "user@email.com",
  "password": "Password@123",
  "company_name": "Teknowxa"
}
```

Response

```
{
  "success": true,
  "user_id": "uuid"
}
```

Login

POST /auth/login

Request

```
{  
  "email": "user@email.com",  
  "password": "Password@123"  
}
```

Response

```
{  
  "access_token": "jwt",  
  "refresh_token": "jwt"  
}
```

Refresh Token

POST /auth/refresh

Logout

POST /auth/logout

Verify Email

POST /auth/verify-email

Forgot Password

POST /auth/forgot-password

Reset Password

POST /auth/reset-password

MODULE 2: ORGANIZATION APIs

Create Organization

POST /organizations

Request

```
{
  "name": "Teknowxa",
  "industry": "IT Services",
  "timezone": "Asia/Kolkata",
  "language": "en"
}
```

Get Organization

GET /organizations/{id}

Update Organization

PUT /organizations/{id}

MODULE 3: TEAM APIs

Invite Team Member

POST /teams/invite

Request

```
{
  "email": "team@email.com",
  "role": "editor"
}
```

Team List

GET /teams

Update Role

PUT /teams/{id}

Remove Team Member

DELETE /teams/{id}

MODULE 4: BRAND KIT APIs

Save Brand Kit

POST /brand-kit

Request

```
{
  "primary_color": "#2563EB",
  "secondary_color": "#F97316",
  "font": "Inter"
}
```

Upload Logo

POST /brand-kit/logo

multipart/form-data

MODULE 5: SOCIAL CONNECT APIs

Connect Facebook

GET /social/facebook/connect

Facebook Callback

GET /social/facebook/callback

Connect Instagram

GET /social/instagram/connect

Connect LinkedIn

GET /social/linkedin/connect

Connect Threads

GET /social/threads/connect

Get Connected Accounts

GET /social/accounts

Response

```
[
  {
    "platform": "facebook",
    "connected": true
  }
]
```

Disconnect Account

DELETE /social/accounts/{id}

MODULE 6: KEYWORD ENGINE APIs

Create Keyword

POST /keywords

Request

```
{
  "keyword": "CRM Software",
  "industry": "IT Services",
}
```

```
{
  "language": "English"
}
```

List Keywords

GET /keywords

MODULE 7: AI CONTENT ENGINE APIs

Generate Caption

POST /ai/generate-caption

Request

```
{
  "keyword": "CRM Software",
  "platform": "linkedin",
  "tone": "professional",
  "language": "English"
}
```

Response

```
{
  "caption": "Manage leads smarter with our CRM solution."
}
```

Generate Hashtags

POST /ai/generate-hashtags

Response

```
{
  "hashtags": [
    "#crm",
    "#automation",
    "#businessgrowth"
  ]
}
```

Generate SEO Content

POST /ai/generate-seo

Generate CTA

POST /ai/generate-cta

Generate Carousel Ideas

POST /ai/generate-carousel

Generate Video Ideas

POST /ai/generate-video-ideas

Generate 30 Day Calendar

POST /ai/generate-calendar

Response

```
{  
  "days": []  
}
```

MODULE 8: MEDIA APIs

Upload Media

POST /media/upload

multipart/form-data

Media Library

GET /media

Delete Media

DELETE /media/{id}

Generate AI Image Prompt

POST /ai/image-prompt

MODULE 9: POSTS APIs

Create Post

POST /posts

Request

```
{
  "title": "CRM Benefits",
  "caption": "...",
  "image_url": "...",
  "platforms": [
    "facebook",
    "linkedin"
  ]
}
```

Update Post

PUT /posts/{id}

Delete Post

DELETE /posts/{id}

Get Post

GET /posts/{id}

List Posts

GET /posts

Filters:

?status=scheduled
?platform=linkedin

MODULE 10: SCHEDULER APIs

Schedule Post

POST /scheduler

Request

```
{  
  "post_id": "uuid",  
  "schedule_time": "2026-06-10T18:00:00Z"  
}
```

Reschedule

PUT /scheduler/{id}

Cancel Schedule

DELETE /scheduler/{id}

Scheduled Posts

GET /scheduler

MODULE 11: PUBLISHING APIs

Publish Now

POST /publish/{post_id}

Flow:

```
Create Queue Job
↓
Publish Worker
↓
Platform Provider
↓
Save Response
```

Retry Failed Post

POST /publish/retry/{post_id}

Publishing Logs

GET /publish/logs/{post_id}

MODULE 12: ANALYTICS APIs

Dashboard Summary

GET /analytics/summary

Response

```
{
  "reach":50000,
  "clicks":1500,
  "engagement":12.5
}
```

Platform Analytics

GET /analytics/platforms

Post Analytics

GET /analytics/posts/{id}

Follower Growth

GET /analytics/followers

Best Posting Time

GET /analytics/best-time

MODULE 13: COMPETITOR APIs

Add Competitor

POST /competitors

Request

```
{
  "platform": "linkedin",
  "profile_url": "https://linkedin.com/company/example"
}
```

Competitor List

GET /competitors

Competitor Analytics

GET /competitors/{id}/analytics

MODULE 14: FESTIVAL APIs

Upcoming Festivals

GET /festivals

Festival Templates

GET /festivals/templates

Auto Generate Festival Post

POST /festivals/generate

MODULE 15: NOTIFICATION APIs

Notifications

GET /notifications

Mark Read

PUT /notifications/{id}/read

Notification Settings

PUT /notifications/settings

MODULE 16: BILLING APIs

Plans

GET /plans

Subscribe

POST /subscriptions

Current Subscription

GET /subscriptions/current

Cancel Subscription

DELETE /subscriptions/current

Payment History

GET /payments

MODULE 17: CLIENT MANAGEMENT APIs

Create Client

POST /clients

Client List

GET /clients

Client Details

GET /clients/{id}

Delete Client

DELETE /clients/{id}

MODULE 18: ADMIN APIs

(Admin Only)

Users

GET /admin/users

Organizations

GET /admin/organizations

Revenue

GET /admin/revenue

Plans

GET /admin/plans
POST /admin/plans
PUT /admin/plans/{id}

Feature Flags

GET /admin/features
PUT /admin/features/{id}

AI Usage

GET /admin/ai-usage

MODULE 19: WEBHOOK APIs

Stripe Webhook

POST /webhooks/stripe

Razorpay Webhook

POST /webhooks/razorpay

Facebook Webhook

POST /webhooks/facebook

Instagram Webhook

POST /webhooks/instagram

LinkedIn Webhook

POST /webhooks/linkedin

MODULE 20: INTERNAL WORKER APIs

(Not Public)

Publish Worker

POST /internal/jobs/publish

Analytics Sync Worker

POST /internal/jobs/analytics-sync

Refresh Token Worker

POST /internal/jobs/token-refresh

Festival Worker

POST /internal/jobs/festival-post

AI Worker

API SECURITY REQUIREMENTS

Mandatory:

JWT Authentication
Role Based Access Control (RBAC)
Rate Limiting
IP Logging
Request Logging
Audit Logging
Webhook Signature Validation
Encrypted OAuth Tokens

SmartAuto Post

Part-4: Backend Folder Structure & Architecture Document

Version: 1.0

Technology Stack:

FastAPI
PostgreSQL
Redis
Celery
AWS S3
Docker
OpenAI

Purpose:

Ye document developer ko batayega:

Project structure
Folder organization
Code standards
Service architecture
Dependency injection
Celery jobs
Provider system
Logging
Error handling

1. PROJECT ARCHITECTURE

Architecture Style:

Modular Monolith

Initial version me microservices mat banao.

Start with:

Single FastAPI Application

Future me modules split kar sakte ho.

2. ROOT STRUCTURE

backend/

```
|— app/  
|— tests/  
|— scripts/  
|— docs/  
|— docker/  
|— migrations/  
|— requirements/  
|— .env  
|— .env.example  
|— docker-compose.yml  
|— alembic.ini  
|— README.md
```

3. APP STRUCTURE

app/

```
|— api/  
|— core/  
|— database/  
|— modules/  
|— services/  
|— providers/  
|— workers/  
|— integrations/  
|— events/  
|— middleware/
```

```
|— schemas/  
|— utils/  
|— storage/  
|— logs/  
|— main.py
```

4. API LAYER

Purpose:

Request Handling
Validation
Response Formatting

Structure:

api/

```
|— v1/  
|   |— auth.py  
|   |— users.py  
|   |— organizations.py  
|   |— social.py  
|   |— content.py  
|   |— posts.py  
|   |— scheduler.py  
|   |— analytics.py  
|   |— billing.py  
|   |— admin.py
```

Controllers should only:

Receive request
Validate
Call service
Return response

Business logic nahi.

5. CORE FOLDER

Application configuration

core/

```
|— config.py
```

```
|— security.py
|— constants.py
|— permissions.py
|— settings.py
|— logger.py
```

6. DATABASE LAYER

Structure:

database/

```
|— session.py
|— base.py
|— repositories/
|
|— models/
|   |— user.py
|   |— organization.py
|   |— post.py
|   |— analytics.py
|
|— migrations/
```

7. MODULE STRUCTURE

Every module should be isolated.

Example:

modules/

```
|— auth/
|— users/
|— organizations/
|— social_accounts/
|— content_ai/
|— posts/
|— analytics/
|— billing/
|— competitors/
|— festivals/
|— notifications/
```

Example Module

```
modules/posts/
```

```
|— router.py  
|— service.py  
|— repository.py  
|— schema.py  
|— model.py  
|— constants.py
```

8. SERVICE LAYER

All business logic here.

```
services/
```

```
|— auth_service.py  
|— post_service.py  
|— content_service.py  
|— analytics_service.py  
|— billing_service.py
```

Example:

```
PostService
```

Responsible for:

```
create post  
update post  
schedule post  
publish post
```

9. REPOSITORY LAYER

Never access DB directly from service.

Structure:

```
repository/
```

```
|— user_repository.py  
|— post_repository.py  
|— analytics_repository.py
```

Flow:

```
API
```

↓
Service
↓
Repository
↓
Database

10. PROVIDER ARCHITECTURE

Critical for social platforms.

Structure:

```
providers/  
├── social/  
│   ├── base_provider.py  
│   ├── facebook_provider.py  
│   ├── instagram_provider.py  
│   ├── linkedin_provider.py  
│   ├── threads_provider.py  
│   └── google_business_provider.py
```

Every provider implements:

```
connect()  
  
publish_post()  
  
delete_post()  
  
fetch_analytics()  
  
refresh_token()
```

This allows:

Future platforms add easily

11. AI ENGINE STRUCTURE

Most important module.

modules/content_ai/

```
├── router.py
├── service.py
├── repository.py
├── generators/
│   ├── caption_generator.py
│   ├── hashtag_generator.py
│   ├── seo_generator.py
│   ├── calendar_generator.py
│   └── cta_generator.py
├── prompts/
│   ├── caption_prompt.py
│   └── seo_prompt.py
├── llm/
│   ├── openai_client.py
│   └── fallback_client.py
```

12. CELERY WORKERS

Structure:

workers/

```
├── celery_app.py
├── tasks/
│   ├── publish_post.py
│   ├── analytics_sync.py
│   ├── refresh_tokens.py
│   ├── generate_content.py
│   └── festival_posts.py
```

Worker Types

Publish Worker

Publish social posts

Analytics Worker

Fetch engagement

AI Worker

Generate captions
Generate hashtags
Generate SEO

Token Worker

Refresh OAuth tokens

13. SCHEDULER STRUCTURE

Use:

Celery Beat

Folder:

workers/schedules/

Jobs:

Publish Posts

Refresh Tokens

Analytics Sync

Festival Generator

Subscription Checker

14. INTEGRATIONS FOLDER

Third-party systems

integrations/

- |— openai/
- |— stripe/
- |— razorpay/
- |— meta/
- |— linkedin/
- |— google_business/

```
|— aws_s3/
```

Never call APIs directly.

Wrap everything.

15. EVENT SYSTEM

Internal communication.

Structure:

```
events/
```

```
|— post_created.py  
|— post_published.py  
|— user_registered.py  
|— payment_success.py
```

Benefits:

Loose coupling

Scalable architecture

16. MIDDLEWARE

```
middleware/
```

```
|— auth.py  
|— organization.py  
|— request_logger.py  
|— rate_limiter.py
```

17. SCHEMAS

Pydantic Models

```
schemas/
```

```
|— auth.py  
|— user.py  
|— post.py  
|— analytics.py
```

18. STORAGE LAYER

storage/

```
|— uploads/  
|— temp/  
|— exports/
```

Production:

AWS S3
Cloudflare R2

19. LOGGING SYSTEM

Structure:

logs/

```
|— app.log  
|— api.log  
|— worker.log  
|— auth.log  
|— scheduler.log
```

Use:

structlog

or

loguru

20. ERROR HANDLING

Global handler

utils/exceptions.py

Custom exceptions:

ValidationException

AuthenticationException

PlatformException

AIException

PaymentException

21. RESPONSE FORMAT

Success:

```
{
  "success": true,
  "message": "Success",
  "data": {}
}
```

Error:

```
{
  "success": false,
  "message": "Validation failed",
  "errors": []
}
```

Entire platform same format.

22. CONFIG MANAGEMENT

Environment files:

```
.env
.env.production
.env.staging
```

Variables:

```
DATABASE_URL
REDIS_URL
OPENAI_KEY
```

S3_KEY

JWT_SECRET

META_APP_ID

META_APP_SECRET

23. TESTING STRUCTURE

tests/

- unit/
- integration/
- api/
- workers/

Coverage Goal:

80%+

24. DOCKER STRUCTURE

docker/

- api/
- postgres/
- redis/
- celery/

docker-compose:

api
postgres
redis
celery_worker
celery_beat
flower

25. DEPLOYMENT STRUCTURE

Stage-1:

1 VPS

FastAPI
PostgreSQL
Redis
Celery

Stage-2:

App Server
DB Server
Worker Server

Stage-3:

Load Balancer
FastAPI Cluster
Redis Cluster
PostgreSQL Replica
Dedicated Workers

26. SECURITY ARCHITECTURE

Must Implement:

JWT Authentication
Refresh Tokens
RBAC
Rate Limiting
Request Logging
Audit Logs
Encrypted OAuth Tokens
IP Tracking
2FA

27. FINAL REQUEST FLOW

Frontend
↓
API Route
↓
Service
↓
Repository
↓
Database

OR

Service
↓
Queue Job
↓
Worker
↓
Provider
↓
Social Platform

SmartAuto Post

Part-5: AI Engine Architecture Document

Version: 1.0

Purpose:

Ye document SmartAuto Post ke sabse important component ko define karta hai:

AI Intelligence Layer

Ye layer hi platform ko Buffer, Hootsuite, SocialPilot jaise tools se alag banayegi.

1. AI ENGINE GOAL

Current tools:

User content likhta hai
User hashtags dhundta hai
User calendar banata hai

User schedule karta hai

SmartAuto Post:

Keyword do
↓
AI samjhe
↓
Content banaye
↓
Hashtags banaye
↓
SEO banaye
↓
Image idea de
↓
Calendar banaye
↓
Schedule kare
↓
Optimize kare

2. AI ARCHITECTURE OVERVIEW

Keyword Input
↓
Keyword Intelligence Engine
↓
Content Brain
↓
Content Generators
↓
Calendar Engine
↓
Optimization Engine
↓
Analytics Feedback Loop

3. AI MODULES

Main AI Modules:

Keyword Intelligence Engine
Caption Generator
Hashtag Generator
CTA Generator
SEO Generator
Calendar Generator
Image Prompt Generator

Trend Engine
Competitor Engine
Optimization Engine

4. KEYWORD INTELLIGENCE ENGINE

Purpose:

User keyword ko samajhna.

Input:

```
{  
  "keyword": "CRM Software"  
}
```

AI Detect Kare:

Industry
Intent
Audience
Pain Points
Use Cases
Related Topics

Output:

```
{  
  "industry": "IT",  
  "audience": "Businesses",  
  "topics": [  
    "Lead Management",  
    "Sales Automation",  
    "Customer Tracking"  
  ]  
}
```

5. CONTENT BRAIN

Ye entire AI system ka center hoga.

Input:

Keyword
Industry
Platform
Tone

Language
Audience

Output:

Content Plan

Example:

CRM Software

Generate:

Benefits Post
Comparison Post
Problem Solution Post
Case Study Post
Testimonial Post

6. CAPTION GENERATOR

Purpose:

Platform specific captions.

Platforms:

Facebook
Instagram
LinkedIn
Threads
Google Business

LinkedIn Structure:

Hook
Problem
Solution
CTA

Instagram Structure:

Short Hook
Emojis
Hashtags
CTA

Facebook Structure:

Story
Value
CTA

Output Example

Still managing leads manually?

Our CRM automates follow-ups, tracks customers and improves conversions.

Book a free demo today.

7. HASHTAG ENGINE

Purpose:

Generate optimized hashtags.

Types:

Primary
Industry
Trending
Location Based

Output:

#crm
#crmsoftware
#businessautomation
#startupindia
#salesgrowth

8. CTA GENERATOR

Generate:

Book Demo
Contact Now
Start Free Trial
Schedule Meeting
Get Quote

Industry-specific.

Example:

School

Apply Now
Admissions Open

Hospital

Book Appointment

9. SEO ENGINE

Generate:

Meta Description
Keywords
LinkedIn Article Intro
Google Business Description

Input:

Website Development

Output:

Professional website development services helping businesses build fast, responsive and SEO optimized websites.

10. IMAGE PROMPT ENGINE

Purpose:

Generate prompts for image creation.

Input:

CRM Software

Output:

Modern CRM dashboard interface with blue gradients, analytics charts, futuristic SaaS design, business growth visualization

Used for:

DALL-E
Stable Diffusion
Flux
Midjourney

11. CALENDAR ENGINE

Purpose:

Generate content plan.

Input:

```
{  
  "industry": "IT",  
  "frequency": "daily"  
}
```

Output:

```
Day 1 Service Highlight  
Day 2 Testimonial  
Day 3 Case Study  
Day 4 Educational  
Day 5 Product Feature  
Day 6 Customer Story  
Day 7 Branding
```

12. FESTIVAL AI ENGINE

Database:

Diwali
Holi
Republic Day
Independence Day

Input:

```
Industry = Hospital  
Festival = Diwali
```

Output:

Happy Diwali from ABC Hospital.
Stay safe and celebrate responsibly.

Plus:

Hashtags
Image Prompt
CTA

13. TREND ENGINE

Purpose:

Detect trending topics.

Sources:

Google Trends
Social Trends
Industry Trends

Store:

trend_keywords
trend_score
trend_category

Output:

Suggested Trending Topics

14. COMPETITOR AI ENGINE

Input:

Competitor LinkedIn Page

Analyze:

Posting Frequency
Best Hashtags
Best Content Type
Engagement Pattern

Output:

Competitor posts videos 4x/week
Most engagement on case studies

15. OPTIMIZATION ENGINE

Purpose:

Analytics se learning.

Input:

Post Performance

Example:

Likes
Comments
Shares
Clicks

Output:

Post more carousels
Reduce hashtags
Best posting time 7 PM

16. PROMPT MANAGEMENT SYSTEM

Folder:

content_ai/prompts/

Files:

caption_prompt.py
hashtag_prompt.py
seo_prompt.py
calendar_prompt.py

Never hardcode prompts.

17. PROMPT VERSIONING

Table:

prompt_templates

Columns:

id
name
version
prompt_text
status

Allows:

A/B Testing
Prompt Upgrades

18. AI MODEL LAYER

Structure:

llm/

openai_client.py
claude_client.py
gemini_client.py
fallback_client.py

Use Provider Pattern.

19. MODEL ROUTER

Logic:

Caption → GPT
SEO → GPT
Calendar → GPT
Quick Hashtags → Local Model

Reduces cost.

20. AI COST OPTIMIZATION

Very Important.

Never send every request to expensive models.

Use:

Templates
+
Rules
+
AI Enhancement

Example:

Generate basic caption
↓
AI improve

Cost drops dramatically.

21. AI CACHE SYSTEM

Redis Cache

Key Example:

caption:crm-software:linkedin

Store:

Generated output

Cache TTL:

24 Hours

Benefits:

Fast response
Lower AI cost

22. AI QUEUE SYSTEM

Heavy AI requests:

30 Day Calendar
Competitor Analysis
Bulk Content Generation

Run via Celery.

Flow:

```
Request
↓
Queue
↓
Worker
↓
Result
```

23. BULK CONTENT GENERATOR

Input:

```
{
  "keyword": "CRM Software",
  "posts": 30
}
```

Output:

30 Captions
30 Hashtag Sets
30 CTA Sets
30 Image Prompts
30 Calendar Entries

One-click month planning.

24. MULTI-LANGUAGE ENGINE

Supported:

English

Hindi
Hinglish
Arabic
French
Spanish

Workflow:

Generate Base Content
↓
Translate
↓
Optimize

25. AI QUALITY SCORING

Every generated content gets:

Readability Score
SEO Score
Engagement Score
Platform Fit Score

Stored in DB.

26. CONTENT SAFETY FILTER

Before publish:

Check:

Spam
Profanity
Duplicate Content
Platform Violations

Block risky content.

27. FUTURE AI AGENTS (V2)

Agent 1:

Content Strategist

Agent 2:

SEO Expert

Agent 3:

Social Media Manager

Agent 4:

Competitor Analyst

Multi-agent workflow:

```
Keyword
↓
Strategist Agent
↓
Content Agent
↓
SEO Agent
↓
Publisher Agent
```

28. AI RESPONSE FORMAT STANDARD

Every AI module returns:

```
{
  "success": true,
  "content": "...",
  "score": 92,
  "tokens_used": 450,
  "generation_time": 1.4
}
```

29. AI PERFORMANCE TARGETS

Caption:

< 1 second

Hashtags:

< 500 ms

SEO:

< 2 sec

Calendar:

< 5 sec

Competitor Analysis:

Background Job

30. AI MASTER FLOW

Keyword
↓
Keyword Intelligence
↓
Content Brain
↓
Caption Engine
↓
Hashtag Engine
↓
SEO Engine
↓
Image Prompt Engine
↓
Calendar Engine
↓
Quality Score
↓
Cache
↓
Response

SmartAuto Post

Part-6: DevOps, Infrastructure & Production Deployment Architecture

Version: 1.0

Purpose:

Ye document define karega ki system ko production me kaise deploy, scale, monitor aur secure karna hai.

Target:

100 Users

↓

1,000 Users

↓

10,000 Users

↓

100,000+ Users

Without rewriting architecture.

1. INFRASTRUCTURE PHILOSOPHY

Goals:

High Availability

High Performance

Scalable

Secure

Fault Tolerant

Cloud Ready

Cost Optimized

2. ENVIRONMENTS

Create 4 environments:

Local

Development

Staging

Production

Structure:

dev.smartautopost.com

staging.smartautopost.com

app.smartautopost.com

3. CLOUD PROVIDER

Recommended:

Phase 1

Contabo VPS
Hetzner Cloud
DigitalOcean

Phase 2

AWS

Recommended Long Term:

AWS

4. PRODUCTION INFRASTRUCTURE

Stage 1:

Internet
↓
Cloudflare
↓
Nginx
↓
FastAPI
↓
PostgreSQL
↓
Redis
↓
Celery

5. DOCKER ARCHITECTURE

Every service containerized.

Containers:

api

postgres

```
redis
```

```
celery_worker
```

```
celery_beat
```

```
flower
```

```
nginx
```

Docker Compose:

```
api
```

```
postgres
```

```
redis
```

```
worker
```

```
beat
```

```
flower
```

```
nginx
```

6. SERVER SPECIFICATION

MVP

8 CPU

16 GB RAM

200 GB SSD

5,000 Users

16 CPU

32 GB RAM

500 GB NVMe

50,000+ Users

App Servers x2

DB Server x1

Redis Server x1

Worker Server x2

7. NGINX CONFIGURATION

Responsibilities:

SSL
Compression
Reverse Proxy
Rate Limiting
Caching

Enable:

gzip
brotli
http2

8. DATABASE ARCHITECTURE

Database:

PostgreSQL 16

Structure:

Primary DB
Read Replica

Later:

Analytics Replica

PostgreSQL Settings

Enable:

connection pooling
query logging

slow query logs

Use:

PgBouncer

9. REDIS ARCHITECTURE

Use for:

Cache

Queue

Sessions

Rate Limiting

AI Cache

Redis Persistence:

RDB

AOF

Enabled.

10. CELERY INFRASTRUCTURE

Worker Types:

Content Worker

Captions

SEO

Hashtags

Publishing Worker

Facebook

Instagram

LinkedIn

Analytics Worker

Analytics Sync

Notification Worker

Emails

Push

11. CELERY BEAT

Runs:

Every Minute

Jobs:

Publish Scheduled Posts

Refresh Tokens

Fetch Analytics

Festival Posts

Subscription Checks

12. STORAGE SYSTEM

Do NOT store images on server.

Use:

AWS S3

or

Cloudflare R2

Store in DB:

URL only

S3 Folder Structure

uploads/

media/

logos/

generated-images/

exports/

13. CDN

Use:

Cloudflare CDN

Handles:

Caching

DDoS Protection

SSL

WAF

14. SSL

Mandatory.

Use:

Cloudflare SSL

LetsEncrypt

Never allow:

HTTP

Only:

HTTPS

15. MONITORING STACK

Use:

Prometheus

Grafana

Monitor:

CPU

RAM

Redis

PostgreSQL

Queue

API Response Time

16. APPLICATION MONITORING

Use:

Sentry

Track:

Errors

Exceptions

Stack Traces

API Failures

17. LOGGING SYSTEM

Centralized logs.

Use:

Loki

or

ELK Stack

Logs:

API Logs

Worker Logs

Auth Logs

AI Logs

Scheduler Logs

18. BACKUP STRATEGY

Database Backup

Daily:

1 AM

Retention:

30 Days

Store:

AWS S3

Media Backup

Weekly

19. CI/CD PIPELINE

Use:

GitHub Actions

Flow:

```
Push Code
↓
Run Tests
↓
Build Docker Image
↓
Deploy Staging
↓
Approval
↓
Deploy Production
```

20. GITHUB STRUCTURE

Repositories:

```
frontend
backend
ai-engine
infrastructure
```

Separate repos.

21. BRANCHING STRATEGY

Branches:

```
main
develop
staging
feature/*
```

22. SECURITY ARCHITECTURE

Mandatory:

JWT

Refresh Tokens

Encrypted Secrets

RBAC

2FA

Audit Logs

IP Tracking

23. SECRET MANAGEMENT

Never store secrets in code.

Use:

AWS Secrets Manager

or

Environment Variables

Store:

OpenAI Key

JWT Secret

DB Password

OAuth Secrets

24. RATE LIMITING

FastAPI Middleware

Limits:

100 requests/minute

Public APIs:

20 requests/minute

25. API GATEWAY (Future)

When scaling:

Cloudflare

↓

API Gateway

↓

Microservices

Gateway Responsibilities:

Auth

Routing

Rate Limits

26. DISASTER RECOVERY

If server dies:

Recovery Plan:

Restore DB Backup

Deploy Docker Stack

Restore Redis

Restore Media

Target:

RTO < 30 minutes

27. HIGH AVAILABILITY ARCHITECTURE

Stage 3:

Cloudflare



Load Balancer

FastAPI #1

FastAPI #2



Redis Cluster



PostgreSQL Primary



Read Replica



Worker Cluster

28. PERFORMANCE TARGETS

API Response:

< 200 ms

Caption Generation:

< 1 sec

Hashtags:

< 500 ms

Scheduler Trigger:

< 5 sec

Dashboard Load:

< 2 sec

29. OBSERVABILITY

Track:

API Success Rate

Queue Length

AI Cost

OpenAI Usage

Failed Posts

Failed OAuth Refreshes

Platform API Errors

Dashboard in Grafana.

30. DEPLOYMENT ROADMAP

Phase 1

Single VPS

Docker

PostgreSQL

Redis

FastAPI

Phase 2

Dedicated DB

Dedicated Worker

Phase 3

Load Balancer

Multiple App Servers

Read Replicas

Phase 4

Kubernetes

Redis Cluster

Autoscaling

Multi Region

SmartAuto Post

Part-7 Complete Development Roadmap & Sprint Planning

Version: 1.0

Purpose:

Ye document define karega:

Project ka exact build order
Developer responsibilities
Sprint planning
Module dependency
Release roadmap
MVP se Production tak ka path

PROJECT DEVELOPMENT PHILOSOPHY

Galti jo startups karte hain:

Sab kuch ek sath banana

Result:

Delays
Bugs
Rewrites
Confusion

SmartAuto Post build order:

Foundation
↓
Auth
↓
Organizations
↓
Social Connect
↓
AI Engine
↓
Posting
↓
Analytics
↓
Billing
↓
Admin
↓
Optimization

TEAM STRUCTURE

Recommended Team

1 Backend Lead
1 AI Engineer
1 Frontend Developer
1 Flutter Developer
1 QA Engineer
1 DevOps Engineer

Minimum Team

1 Backend Developer
1 Frontend Developer
1 Full Stack AI Developer

PHASE 1

FOUNDATION SETUP

Duration:

1 Week

Goal:

System bootstrap

Tasks:

Backend Setup

FastAPI Setup
PostgreSQL Setup
Redis Setup
Celery Setup
Docker Setup
Environment Setup

Repository Setup

GitHub Repositories
Branch Rules
PR Rules
Code Standards

CI/CD Setup

GitHub Actions
Docker Build
Staging Deploy

Deliverables:

Running Backend
Running Database
Running Redis
Working Docker

PHASE 2

AUTHENTICATION SYSTEM

Duration:

1 Week

Build:

Register
Login
JWT Auth
Refresh Token
Email Verification
Forgot Password
Reset Password

Database:

users
user_sessions

API Ready:

/auth/*

Deliverables:

Complete Authentication Module

PHASE 3

ORGANIZATION + TEAM MANAGEMENT

Duration:

1 Week

Build:

Organizations
Roles
Permissions
Team Invite

Team Members

Roles:

Owner
Admin
Manager
Editor
Viewer

Deliverables:

Multi Tenant SaaS Foundation

PHASE 4

BRAND KIT + MEDIA SYSTEM

Duration:

5 Days

Build:

Logo Upload
Brand Colors
Font Selection
Media Upload
Media Library
S3 Integration

Deliverables:

Brand Assets System

PHASE 5

SOCIAL CONNECTION ENGINE

Duration:

2 Weeks

Build:

Facebook

OAuth
Token Storage
Refresh Token

LinkedIn

OAuth
Posting Permission

Instagram

Business Account Integration

Database:

social_accounts

Deliverables:

Connected Accounts Module

PHASE 6

AI ENGINE V1

Duration:

2 Weeks

Build:

Keyword Intelligence

Keyword Analysis
Topic Extraction

Caption Generator

LinkedIn Captions
Facebook Captions
Instagram Captions

Hashtag Generator

CTA Generator

SEO Generator

AI Stack:

OpenAI
Prompt Templates
Redis Cache

Deliverables:

Content Generation Engine

PHASE 7

POST MANAGEMENT SYSTEM

Duration:

1 Week

Build:

Create Post
Edit Post
Delete Post
Drafts
Post History

Database:

posts
post_platforms

Deliverables:

Post Manager

PHASE 8

CALENDAR & SCHEDULER

Duration:

1 Week

Build:

Schedule Post
Calendar View
Recurring Posts
Queue System

Celery:

Publish Jobs

Deliverables:

Scheduler Engine

PHASE 9

SOCIAL PUBLISHING ENGINE

Duration:

2 Weeks

Build:

Facebook Publishing

LinkedIn Publishing

Instagram Publishing

Threads Publishing

Provider Architecture:

Base Provider
Facebook Provider
Instagram Provider

LinkedIn Provider
Threads Provider

Deliverables:

Auto Posting System

PHASE 10

ANALYTICS ENGINE

Duration:

2 Weeks

Collect:

Likes
Comments
Shares
Reach
Clicks
Followers

Build:

Analytics Dashboard
Platform Comparison
Best Posting Time

Deliverables:

Analytics Module

PHASE 11

AI CALENDAR GENERATOR

Duration:

1 Week

Generate:

7 Day Calendar
30 Day Calendar
90 Day Calendar

Output:

Post Ideas
Captions
Hashtags

Deliverables:

Calendar AI

PHASE 12

FESTIVAL AUTOMATION

Duration:

5 Days

Build:

Festival Database
Auto Festival Posts
Festival Templates

Deliverables:

Festival Engine

PHASE 13

CLIENT MANAGEMENT

Duration:

1 Week

Build:

Client Accounts

Client Analytics
Client Posts
Client Social Accounts

Agency Features.

Deliverables:

Agency Module

PHASE 14

BILLING SYSTEM

Duration:

1 Week

Integrations:

Razorpay
Stripe

Build:

Plans
Subscriptions
Invoices
Usage Limits

Deliverables:

Monetization Ready

PHASE 15

ADMIN PANEL

Duration:

1 Week

Build:

User Management
Organization Management
Plans
Revenue
Feature Flags
AI Usage

Deliverables:

Complete Admin Panel

PHASE 16

COMPETITOR ANALYSIS

Duration:

2 Weeks

Build:

Competitor Tracking
Hashtag Analysis
Content Analysis
Posting Frequency

AI Suggestions.

Deliverables:

Competitor Engine

PHASE 17

AI OPTIMIZATION ENGINE

Duration:

2 Weeks

Analyze:

Analytics

Engagement
Platform Performance

Suggest:

Best Time
Best Content
Best Hashtags

Deliverables:

Optimization AI

PHASE 18

MOBILE APP APIs

Duration:

1 Week

Build:

Mobile API Layer
Push Notifications
Device Management

Deliverables:

Flutter Ready APIs

PHASE 19

SECURITY HARDENING

Duration:

1 Week

Implement:

Rate Limiting
2FA

Audit Logs
OAuth Encryption
WAF Rules

Deliverables:

Production Security

PHASE 20

LOAD TESTING

Duration:

5 Days

Tools:

Locust
k6

Test:

1,000 Users

5,000 Users

10,000 Users

Deliverables:

Performance Report

MVP RELEASE (Version 1.0)

Must Include:

Auth
Organizations
Teams
Brand Kit
Facebook Connect
LinkedIn Connect
AI Captions
AI Hashtags

Posts
Scheduler
Analytics
Billing
Admin Panel

Expected Timeline:

10-12 Weeks

VERSION 2.0

Add:

Instagram Advanced Features
Threads
Festival Automation
Agency Features
Competitor Tracking
AI Calendar

Timeline:

4-6 Weeks

VERSION 3.0

Add:

WhatsApp Automation
AI Agents
Trend Engine
Advanced Analytics
Multi Language AI

Timeline:

6-8 Weeks

RELEASE ORDER (MOST IMPORTANT)

Build in this exact order:

Foundation

↓
Auth
↓
Organizations
↓
Brand Kit
↓
Social Connect
↓
AI Content
↓
Posts
↓
Scheduler
↓
Publishing
↓
Analytics
↓
Billing
↓
Admin
↓
Competitor AI
↓
Optimization AI

SmartAuto Post

Part-8 Social Platform Integration Specification

Version: 1.0

Purpose:

Ye document define karega:

Facebook Integration
Instagram Integration
LinkedIn Integration
Threads Integration
Google Business Integration
OAuth Flow
Publishing Flow
Analytics Flow
Token Management
Rate Limits
Failure Handling

Ye SmartAuto Post ka sabse critical backend document hai.

1. SOCIAL INTEGRATION ARCHITECTURE

```
Frontend
  ↓
FastAPI Backend
  ↓
Social Provider Layer
  ↓
Platform APIs

Facebook
Instagram
LinkedIn
Threads
Google Business
```

2. PROVIDER DESIGN PATTERN

Sab platforms ek hi interface implement karenge.

```
class SocialProvider:

    connect()

    refresh_token()

    publish_post()

    delete_post()

    get_post()

    get_analytics()

    disconnect()
```

Folder Structure

```
providers/social/

facebook_provider.py
```

```
instagram_provider.py
linkedin_provider.py
threads_provider.py
google_business_provider.py
```

3. FACEBOOK INTEGRATION

API

Meta Graph API

Requirements

User must have:

Facebook Account

Facebook Page

Page Admin Access

OAuth Permissions

pages_show_list

pages_manage_posts

pages_read_engagement

business_management

Connection Flow

Connect Facebook

↓

Meta Login

↓

User Approves

↓

Authorization Code

↓

Access Token
↓
Save Database

Database

social_accounts

platform = facebook

page_id

page_name

access_token

expires_at

Publish Flow

Scheduled Job
↓
Facebook Provider
↓
Graph API
↓
Publish
↓
Store Response

Supported Content

Text

Image

Link

Carousel (Future)

Analytics

Fetch:

Reach

Impressions

Likes

Comments

Shares

4. INSTAGRAM INTEGRATION

Important:

Instagram Personal Accounts unsupported.

Must use:

Instagram Business Account

Connected to:

Facebook Page

Permissions

instagram_basic

instagram_content_publish

pages_show_list

Publishing Flow

Instagram requires:

Step 1

Create Media Container

Step 2

Publish Media Container

Supported

Image Posts

Carousel Posts

Video Posts

Analytics

Reach

Likes

Comments

Saves

Impressions

5. LINKEDIN INTEGRATION

API

LinkedIn Marketing Developer Platform

OAuth Scopes

openid

profile

email

w_member_social

Supported

Text Post

Image Post

Article Share

Company Page Post

Publishing Flow

Create Asset
↓
Upload Image
↓
Create Post
↓
Publish

Analytics

Likes

Comments

Shares

Impressions

Clicks

6. THREADS INTEGRATION

API

Meta Threads API

Requirements

Threads Account

Meta Business App

Supported

Text Posts

Image Posts

Future

Thread Series

Conversation Chains

7. GOOGLE BUSINESS PROFILE

Very powerful for local businesses.

API

Google Business Profile API

OAuth Scopes

business.manage

Supported

Offers

Updates

Events

Photos

Analytics

Views

Clicks

Calls

Direction Requests

8. TOKEN MANAGEMENT SYSTEM

Critical Module.

Access Token Storage

Never plain text.

Store encrypted.

AES Encryption

Refresh Strategy

Cron:

Every 6 Hours

Run:

RefreshTokenJob

Expiry Alerts

Notify user:

Account Needs Reconnection

9. SOCIAL ACCOUNT STATES

Possible states:

connected

expired

revoked

failed

reconnecting

10. PUBLISHING ENGINE FLOW

User Creates Post
↓
Schedule Created
↓
Celery Job
↓
Provider Selected
↓
API Request
↓
Platform Response
↓
Save Result

11. POST STATUS FLOW

draft
↓
scheduled
↓
processing
↓
published

or

failed

12. FAILURE RETRY SYSTEM

If API fails:

Retry 1

after 1 minute

Retry 2

after 5 minutes

Retry 3

after 15 minutes

Then:

Mark Failed

Notify User

13. ANALYTICS COLLECTION FLOW

Cron:

Every 6 Hours

Process

Get Published Posts

↓

Fetch Analytics

↓

Store Metrics

↓

Update Dashboard

14. WEBHOOK SYSTEM

Future Ready.

Platforms:

Facebook

Instagram

LinkedIn

Events:

Comment

Mention

Post Update

Engagement

Store:

webhook_logs

15. RATE LIMIT STRATEGY

Every platform has limits.

Create:

```
rate_limit_tracker
```

Table:

```
platform  
  
remaining_calls  
  
reset_time
```

Before API Call:

Check:

Limit Available?

If No:

Queue Delay

16. PROVIDER RESPONSE STANDARD

Every provider returns:

```
{  
  "success": true,  
  "platform": "facebook",  
  "platform_post_id": "123456",  
  "message": "Published Successfully"  
}
```

Failure:

```
{  
  "success": false,  
  "error_code": "TOKEN_EXPIRED",  
}
```

```
"message": "Reconnect Account"
}
```

17. SOCIAL LOGGING SYSTEM

Table:

social_logs

Fields:

id

platform

request

response

status

created_at

18. SOCIAL HEALTH DASHBOARD

Admin Panel:

Show:

Facebook API Status

Instagram API Status

LinkedIn API Status

Threads API Status

Monitor:

Failures

Token Expiry

API Downtime

19. FUTURE INTEGRATIONS (V2)

Planned:

Twitter/X

Pinterest

YouTube Community

TikTok

Telegram Channels

20. PLATFORM SUPPORT MATRIX

Feature	FB	IG	LinkedIn	Threads	GBP
Text Post	✓	✓	✓	✓	✓
Image Post	✓	✓	✓	✓	✓
Video Post	✓	✓	✓	Planned	✗
Carousel	✓	✓	✗	✗	✗
Analytics	✓	✓	✓	Limited	✓
OAuth	✓	✓	✓	✓	✓
Scheduling	✓	✓	✓	✓	✓

SmartAuto Post

Part-9 Frontend Architecture Document

Version: 1.0

Purpose:

Ye document define karega:

Web Architecture

Mobile Architecture

UI System

State Management

API Layer

Authentication Flow

Dashboard System
Responsive Design
Performance Optimization

Frontend Team isi document ke basis par pura system build karegi.

1. FRONTEND TECH STACK

Web

Next.js 15
TypeScript
Tailwind CSS
ShadCN UI
React Query
Zustand
React Hook Form
Zod

Mobile

FlutterFlow
Flutter 3.x
Riverpod
GoRouter

2. FRONTEND ARCHITECTURE

Architecture Style:

Feature Based Architecture

Avoid:

pages/
components/
sab ek jagah

Use:

features based modules

3. NEXT.JS PROJECT STRUCTURE

frontend/

src/

- |— app/
- |— components/
- |— features/
- |— services/
- |— hooks/
- |— stores/
- |— lib/
- |— providers/
- |— layouts/
- |— constants/
- |— types/
- |— styles/
- |— assets/

4. APP ROUTER STRUCTURE

app/

- |— login
- |— register
- |— forgot-password
- |— dashboard
- |— posts
- |— content-ai
- |— analytics
- |— scheduler
- |— social-accounts
- |— media-library
- |— competitors
- |— billing
- |— settings
- |— admin

5. FEATURE STRUCTURE

Example:

features/posts/

- |— api/
- |— components/
- |— hooks/
- |— types/
- |— store/
- |— pages/

6. STATE MANAGEMENT

Use:

Zustand

Stores:

authStore

organizationStore

postStore

analyticsStore

notificationStore

themeStore

Example:

authStore

user

token

permissions

organization

7. SERVER STATE MANAGEMENT

Use:

TanStack Query

Handles:

API Caching

Refetching

Optimistic Updates

Background Sync

Example:

```
usePosts()
```

```
useAnalytics()
```

```
useOrganizations()
```

8. API LAYER

Never call API directly.

Create:

```
services/
```

```
api.ts
```

```
auth.service.ts
```

```
post.service.ts
```

```
analytics.service.ts
```

```
billing.service.ts
```

Example:

```
postService.create()
```

```
postService.update()
```

```
postService.schedule()
```

9. AUTHENTICATION FLOW

Login

```
User Login
↓
Receive JWT
↓
Store Access Token
↓
Store Refresh Token
↓
Load User
↓
Redirect Dashboard
```

Refresh Flow

```
Access Token Expired
↓
Refresh Endpoint
↓
New Token
```

10. ROUTE PROTECTION

Public Routes

```
/login
/register
/forgot-password
```

Protected Routes

```
/dashboard
/posts
/analytics
/settings
```

Admin Routes

/admin/*

11. LAYOUT ARCHITECTURE

Main Layout

Sidebar

Top Navbar

Content Area

Right Utility Panel

Structure

Top Navbar	
Side Bar	Content Area

12. SIDEBAR MENU

Dashboard

Create Post

AI Content

Calendar

Posts

Analytics

Connected Accounts

Media Library

Competitors

Clients

Billing

Settings

Admin:

Users

Organizations

Revenue

Plans

Feature Flags

13. DASHBOARD ARCHITECTURE

Widgets:

Total Posts

Scheduled Posts

Published Posts

Followers Growth

Engagement Rate

Top Platform

AI Usage

Subscription Usage

Dashboard Sections

Summary Cards

Charts

Recent Posts

Upcoming Scheduled Posts

AI Suggestions

14. CHART SYSTEM

Library:

Recharts

Charts:

Line Chart

Bar Chart

Area Chart

Pie Chart

Analytics Screens

Engagement Growth

Follower Growth

Reach

Clicks

Platform Comparison

15. POST CREATION SCREEN

Sections:

Keyword Input

Caption Generator

Hashtag Generator

CTA Generator

Image Upload

Platform Selection

Schedule Selection

Flow:

Keyword
↓
Generate Content
↓
Edit
↓
Schedule
↓
Publish

16. AI CONTENT PAGE

Tabs:

Captions

Hashtags

SEO

CTA

Carousel Ideas

Video Ideas

Features:

Copy

Regenerate

Save

Schedule

17. CALENDAR PAGE

Views:

Month

Week

Day

Features:

Drag Drop

Reschedule

Quick Edit

Platform Filters

18. SOCIAL ACCOUNTS PAGE

Cards:

Facebook

Instagram

LinkedIn

Threads

Google Business

Each Card Shows:

Status

Account Name

Last Sync

Reconnect Button

19. MEDIA LIBRARY

Tabs:

Uploaded

Templates

AI Images

Features:

Search

Filter

Delete

Preview

20. ANALYTICS PAGE

Sections:

Overview

Platforms

Posts

Audience

Best Time

Filters:

Today

7 Days

30 Days

90 Days

21. BILLING PAGE

Cards:

Current Plan

Usage

Renewal Date

Upgrade Plan

Tables:

Invoices

Payments

Transactions

22. SETTINGS PAGE

Tabs:

Profile

Organization

Security

Notifications

API Keys

23. ADMIN PANEL STRUCTURE

Pages:

Dashboard

Users

Organizations

Subscriptions

Revenue

AI Usage

Feature Flags

System Health

24. COMPONENT ARCHITECTURE

Shared Components

Button

Input

Modal

Drawer

Dropdown

DataTable

DatePicker

ChartCard

Location:

components/ui/

25. TABLE SYSTEM

Use:

TanStack Table

Features:

Search

Pagination

Sorting

Column Visibility

Bulk Actions

26. FORM SYSTEM

Use:

React Hook Form

Validation:

Zod

Example:

Create Post Form

Schedule Form

Billing Form

27. THEME SYSTEM

Modes:

Light

Dark

Primary Colors

#2563EB

#4F46E5

Accent

#F97316

28. RESPONSIVE SYSTEM

Breakpoints

Mobile

Tablet

Desktop

Wide Screen

Sidebar

Desktop → Fixed

Mobile → Drawer

29. PERFORMANCE OPTIMIZATION

Use:

Lazy Loading

Dynamic Imports

Image Optimization

Caching

Avoid:

Large Bundles

Heavy Re-renders

30. NOTIFICATION SYSTEM

Use:

Sonner Toast

Types:

Success

Warning

Error

Info

31. ERROR HANDLING

Global Error Boundary

Pages:

404

500

32. MOBILE APP STRUCTURE (Flutter)

lib/
features/
core/
services/
widgets/
models/
routes/
providers/

Screens:

Login
Register
Dashboard
Posts
AI Content
Calendar
Analytics
Settings

33. MOBILE STATE MANAGEMENT

Use:

Riverpod

Stores:

Auth

Posts

Analytics

Settings

34. PUSH NOTIFICATIONS

Use:

Firestore Cloud Messaging

Events:

Post Published

Post Failed

Subscription Expiring

Analytics Report Ready

35. FRONTEND SECURITY

Implement:

JWT Handling

Route Guards

Permission Checks

XSS Protection

CSRF Protection

36. FINAL FRONTEND FLOW

Login

↓

Dashboard

↓

Connect Accounts

↓

Create Content

↓
Generate AI Content
↓
Schedule
↓
Publish
↓
Analytics
↓
Optimization Suggestions

SmartAuto Post

Part-10 Product Requirements Document (PRD)

Version: 1.0

Product Type: AI Powered Social Media Automation SaaS

1. PRODUCT VISION

SmartAuto Post ek AI-powered platform hai jo businesses, agencies aur creators ko social media content create, optimize, schedule aur publish karne me help karega.

Goal:

Keyword do
↓
AI Content Generate Kare
↓
Hashtags Generate Kare
↓
Image Ideas Generate Kare
↓
Schedule Kare
↓
Auto Publish Kare
↓
Analytics Track Kare
↓
Optimization Suggestions De

User ko Facebook, Instagram, LinkedIn, Threads manually open karne ki zarurat na pade.

2. TARGET USERS

Small Businesses

Hospitals
Schools
Restaurants
Gyms
Real Estate
Clinics

Agencies

Digital Marketing Agencies
Social Media Agencies
SEO Agencies

Creators

Influencers
Coaches
Consultants
Freelancers

Enterprises

Corporate Brands
Multi-location Businesses

3. USER ROLES

Owner

Permissions:

Everything

Admin

Permissions:

Team Management
Posts
Analytics
Billing View

Manager

Permissions:

Create Posts
Schedule Posts
View Analytics

Editor

Permissions:

Create Content
Edit Content
Schedule Content

Viewer

Permissions:

Read Only

4. USER STORIES

Authentication

US-001

As a user,

I want to register
So that I can use the platform

Acceptance Criteria:

Email required

Password required

Email verification sent

US-002

As a user,

I want to login

So that I can access my dashboard

Acceptance Criteria:

JWT generated

Refresh token generated

5. ORGANIZATION MANAGEMENT

US-003

As an owner,

I want to create an organization

So that I can manage my business

Acceptance:

Organization name required

Timezone required

US-004

As an owner,

I want to invite team members

Acceptance:

Invitation email sent

Role assigned

6. SOCIAL ACCOUNT CONNECTION

US-005

As a user,

I want to connect Facebook

Acceptance:

OAuth successful

Token stored

Page selected

US-006

As a user,

I want to connect LinkedIn

Acceptance:

LinkedIn OAuth completed

US-007

As a user,

I want to connect Instagram Business

Acceptance:

Instagram connected

Business Account verified

7. AI CONTENT GENERATION

US-008

As a marketer,

I want AI captions

Input:

Keyword
Platform
Tone

Output:

Caption

Acceptance:

Generated in under 3 seconds

US-009

As a marketer,

I want AI hashtags

Acceptance:

Minimum 15 hashtags

US-010

As a marketer,

I want AI CTA suggestions

Acceptance:

At least 5 CTAs

US-011

As a marketer,

I want SEO content

Acceptance:

Meta description

Keywords

SEO snippet

8. POST MANAGEMENT

US-012

As a user,

I want to create a post

Fields:

Title

Caption

Media

Platforms

US-013

As a user,

I want to save drafts

Acceptance:

Draft retrievable later

US-014

As a user,

I want to duplicate posts

Acceptance:

New draft created

9. SCHEDULER

US-015

As a user,

I want to schedule a post

Acceptance:

Date required

Time required

Timezone applied

US-016

As a user,

I want recurring posts

Options:

Daily

Weekly

Monthly

10. AUTO PUBLISHING

US-017

As a user,

I want posts automatically published

Acceptance:

Published at scheduled time

US-018

As a user,

I want retry on failure

Acceptance:

3 retries

11. ANALYTICS

US-019

As a user,

I want analytics dashboard

Metrics:

Reach

Impressions

Likes

Comments

Shares

Clicks

US-020

As a user,

I want best posting time suggestions

Acceptance:

AI recommendations displayed

12. CONTENT CALENDAR

US-021

As a user,

I want AI-generated monthly calendar

Acceptance:

30 content ideas generated

13. FESTIVAL AUTOMATION

US-022

As a user,

I want automatic festival posts

Acceptance:

Industry specific content generated

14. COMPETITOR ANALYSIS

US-023

As a user,

I want competitor insights

Output:

Posting frequency

Top hashtags

Top content

15. MEDIA LIBRARY

US-024

As a user,

I want media management

Features:

Upload

Search

Delete

Filter

16. BILLING

US-025

As a user,

I want subscription plans

Plans:

Free

Pro

Agency

Enterprise

US-026

As a user,

I want invoices

Acceptance:

Download PDF

17. NOTIFICATIONS

US-027

As a user,

I want notifications

Events:

Post Published

Post Failed

Plan Expiring

Analytics Ready

18. ADMIN PANEL

US-028

As admin,

I want user management

Actions:

View

Suspend

Delete

US-029

As admin,

I want revenue dashboard

Metrics:

MRR

ARR

Total Revenue

Active Users

19. SUBSCRIPTION RULES

Free Plan

1 Social Account

20 Posts/Month

Basic Analytics

Pro Plan

10 Social Accounts

Unlimited Posts

Advanced Analytics

Agency Plan

50 Clients

Team Members

Client Analytics

Enterprise

Custom Limits

Dedicated Support

Custom Features

20. AI USAGE RULES

Track:

Prompt Tokens

Completion Tokens

Cost

Response Time

Limits:

Per Plan Usage

21. BUSINESS RULES

Rule 1:

User must verify email before publishing

Rule 2:

Social account required before scheduling

Rule 3:

Expired tokens block publishing

Rule 4:

Only Owners can manage billing

Rule 5:

Only Admin can manage feature flags

22. EDGE CASES

Publishing

Case:

Platform API Down

Expected:

Retry Queue

Case:

Token Expired

Expected:

Refresh Token

Else Notify User

Case:

Image Missing

Expected:

Prevent Publish

Case:

Rate Limit Reached

Expected:

Queue Delay

23. NON-FUNCTIONAL REQUIREMENTS

Performance:

API Response < 200ms

AI:

Caption < 3 seconds

Availability:

99.9%

Security:

JWT

2FA

Encrypted Tokens

Audit Logs

24. SUCCESS METRICS

Business KPIs:

Active Users

Posts Published

AI Generations

Retention Rate

MRR

ARR

Product KPIs:

Publishing Success Rate

AI Response Time

Scheduler Accuracy

Platform Connection Success

25. MVP DEFINITION (VERSION 1.0)

Must Have:

Authentication

Organizations

Teams

Facebook

LinkedIn

AI Captions

AI Hashtags

Posts

Scheduler

Publishing

Analytics

Billing

Admin Panel

26. VERSION 2.0

Instagram Advanced

Threads

Festival Automation

AI Calendar

Agency Features

27. VERSION 3.0

AI Agents

Competitor Intelligence

Trend Detection

WhatsApp Automation

Advanced AI Analytics

SmartAuto Post

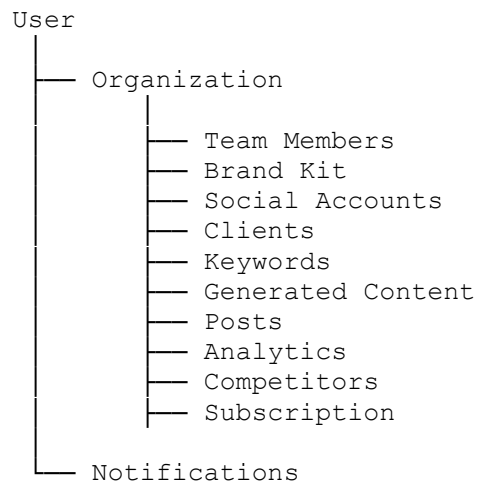
Part-11 Database ERD (Entity Relationship Design)

Version: 1.0

Purpose:

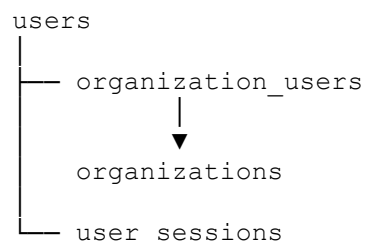
Database relationships
Foreign keys
Multi-tenant architecture
Data ownership rules
Cascade rules
Query optimization

MASTER ERD OVERVIEW



LEVEL 1

AUTH & ORGANIZATION



Relationship:

User
↓
Many Organizations

Organization
↓
Many Users

Tables:

users.id
↓
organization_users.user_id

organizations.id
↓
organization_users.organization_id

LEVEL 2

ORGANIZATION CORE

organizations
|
— brand_kits
— clients
— social_accounts
— keywords
— posts
— competitors
— subscriptions

BRAND KIT RELATIONSHIP

organizations
↓
brand_kits

One-to-One

brand_kits.organization_id
FK organizations.id

CLIENT MANAGEMENT ERD

```
organizations
  |
  ▼
clients
  |
  ▼
client_users
```

Relationship:

One Organization
Many Clients

One Client
Many Users

SOCIAL ACCOUNT ERD

```
organizations
  |
  ▼
social_accounts
```

One organization can connect:

Facebook

Instagram

LinkedIn

Threads

Google Business

Example:

```
Organization
  |
  ├── Facebook Page
  ├── LinkedIn Page
  ├── Instagram Business
  └── Threads Account
```

KEYWORD ENGINE ERD

organizations
↓
keywords
↓
generated_contents

Relationship:

One Keyword
Many Generated Contents

Example:

CRM Software
|
— Caption
— Hashtags
— CTA
— SEO Content

CONTENT AI ERD

keywords
↓
generated_contents
↓
hashtags

Relationship:

Generated Content
|
— Caption
— SEO
— CTA
— Hashtags

MEDIA LIBRARY ERD

organizations
↓
media_assets

One organization

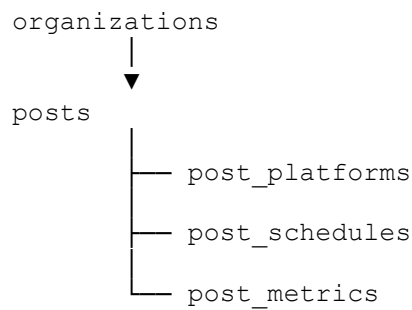
Many Images

Many Videos

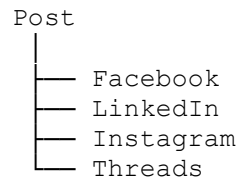
Many Templates

POST MANAGEMENT ERD

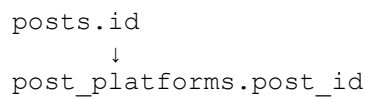
Most important module.



POST RELATIONSHIP



Tables:



PLATFORM RELATIONSHIP



Example:

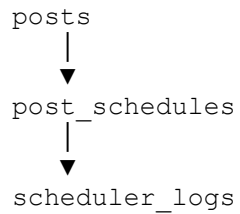
Post #100

Facebook
LinkedIn
Instagram

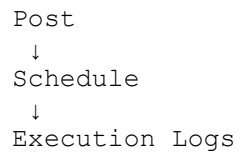
One Post

Many Platform Records

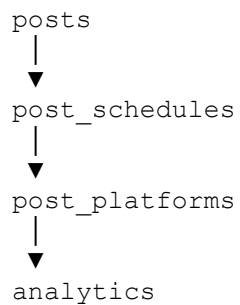
SCHEDULER ERD



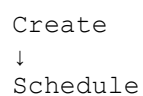
Relationship:



PUBLISHING FLOW ERD



Flow:



↓
Publish
↓
Track

ANALYTICS ERD

posts
↓
post_metrics

social_accounts
↓
follower_metrics

ANALYTICS SUMMARY

organizations
↓
analytics_summary

Stores:

Total Reach

Total Clicks

Engagement

Followers

COMPETITOR ERD

organizations
↓
competitors
↓
competitor_metrics

Relationship:

One Organization

Many Competitors

Example:

Competitor A

Competitor B

Competitor C

FESTIVAL ENGINE ERD

festival_master



festival_templates

Example:

Diwali



Hospital Template

School Template

Real Estate Template

NOTIFICATION ERD

users



notifications

Relationship:

One User

Many Notifications

BILLING ERD

plans



subscriptions
↓
payments

BILLING FLOW

Plan
↓
Subscription
↓
Payment

Example:

Agency Plan
↓
Subscription
↓
Razorpay Payment

AI USAGE ERD

organizations
↓
ai_requests

Track:

Prompt Tokens
Completion Tokens
Cost
Response Time

Used For:

Billing
Analytics
Cost Control

AUDIT SYSTEM ERD

users
|
▼
audit_logs

Stores:

Who

What

When

Old Value

New Value

Example:

User Changed Billing Plan

ADMIN ERD

admin_users
|
▼
feature_flags

Admin controls:

AI Features

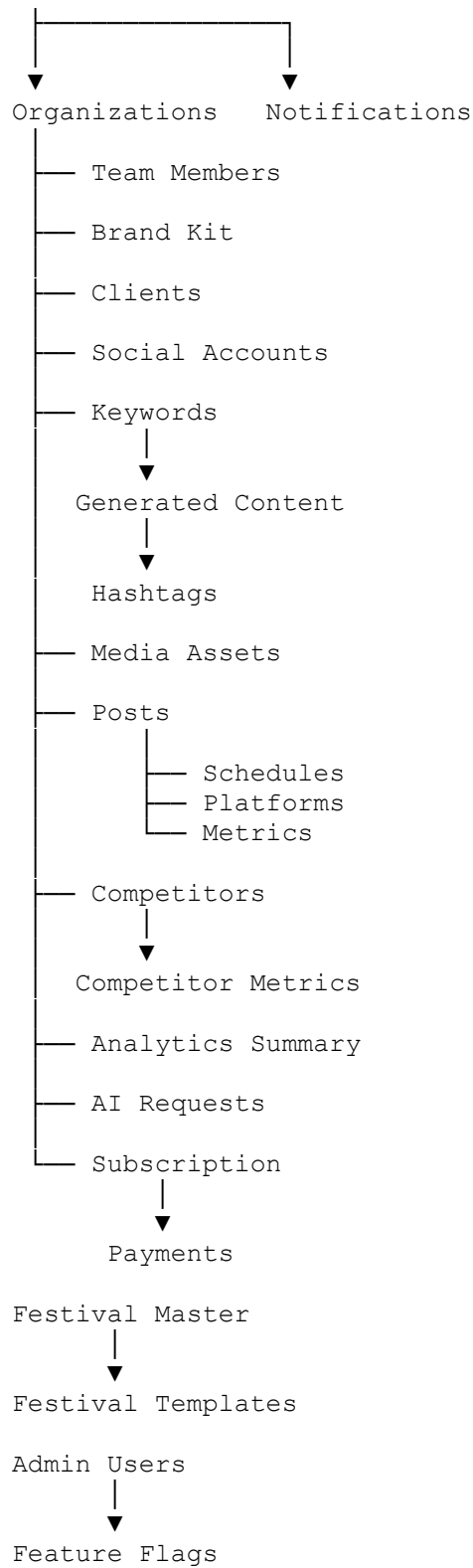
Competitor Module

Festival Automation

WhatsApp Integration

COMPLETE MASTER RELATIONSHIP DIAGRAM

Users
|



CRITICAL FOREIGN KEYS

Must enforce:

organization_users.organization_id
→ organizations.id

organization_users.user_id
→ users.id

social_accounts.organization_id
→ organizations.id

posts.organization_id
→ organizations.id

post_platforms.post_id
→ posts.id

post_schedules.post_id
→ posts.id

post_metrics.post_id
→ posts.id

subscriptions.organization_id
→ organizations.id

payments.subscription_id
→ subscriptions.id

generated_contents.keyword_id
→ keywords.id

hashtags.content_id
→ generated_contents.id

MULTI-TENANT SECURITY RULE

Every business table must contain:

organization_id

Example:

posts

keywords

social_accounts

media_assets

competitors

analytics

Reason:

Organization A data
never visible to
Organization B

DATABASE SCALING STRATEGY

Large tables:

posts

post_metrics

ai_requests

notifications

audit_logs

Future:

Partition by Month

Example:

post_metrics_2026_01

post_metrics_2026_02

DATABASE SIZE ESTIMATION

10,000 Organizations:

Users ~100k

Posts ~50M+

Analytics ~500M+

AI Requests ~100M+

Notifications ~50M+

Isliye analytics aur logs ko future me separate database (ClickHouse) me shift karne ka plan rakha gaya hai.

SmartAuto Post

Part-12 Complete OpenAPI / Swagger Specification

Version: 1.0

Purpose:

Ye document define karega:

Swagger Structure
API Standards
Request Format
Response Format
Error Standards
Pagination Standards
Authentication Scheme
Webhook Contracts
Worker APIs

Ye backend team directly Swagger/OpenAPI generate karne ke liye use karegi.

OPENAPI VERSION

openapi: 3.1.0

Base URL

Production:
<https://api.smartautopost.com/api/v1>

Staging:
<https://staging-api.smartautopost.com/api/v1>

SECURITY SCHEME

JWT Authentication

```
components:
  securitySchemes:
    BearerAuth:
      type: http
      scheme: bearer
      bearerFormat: JWT
```

Protected APIs:

```
security:
  - BearerAuth: []
```

GLOBAL RESPONSE FORMAT

Success:

```
{
  "success": true,
  "message": "Success",
  "data": {}
}
```

Error:

```
{
  "success": false,
  "message": "Validation failed",
  "errors": []
}
```

STANDARD RESPONSE SCHEMAS

SuccessResponse

```
SuccessResponse:
  type: object
  properties:
    success:
      type: boolean
      example: true

    message:
      type: string

    data:
```

```
type: object
```

ErrorResponse

```
ErrorResponse:
  type: object

  properties:
    success:
      type: boolean

    message:
      type: string

    errors:
      type: array
```

PAGINATION STANDARD

Every listing API must return:

```
{
  "success": true,
  "data": {
    "items": [],
    "pagination": {
      "page": 1,
      "per_page": 20,
      "total": 100,
      "total_pages": 5
    }
  }
}
```

COMMON QUERY PARAMETERS

```
page:
  type: integer

per_page:
  type: integer

search:
  type: string

sort_by:
  type: string
```

```
sort_order:  
  type: string
```

ERROR CODE STANDARD

Authentication Errors

401_UNAUTHORIZED

TOKEN_EXPIRED

INVALID_TOKEN

ACCOUNT_DISABLED

Validation Errors

VALIDATION_ERROR

MISSING_FIELD

INVALID_FORMAT

Social Platform Errors

TOKEN_EXPIRED

ACCOUNT_DISCONNECTED

RATE_LIMIT_EXCEEDED

PLATFORM_DOWN

AI Errors

AI_TIMEOUT

AI_PROVIDER_DOWN

AI_QUOTA_EXCEEDED

Billing Errors

PAYMENT_FAILED

SUBSCRIPTION_EXPIRED

PLAN_LIMIT_REACHED

AUTH SCHEMAS

RegisterRequest

RegisterRequest:

type: object

required:

- name
- email
- password

properties:

name:

type: string

email:

type: string

password:

type: string

LoginRequest

LoginRequest:

type: object

required:

- email
- password

properties:

email:

type: string

password:

type: string

LoginResponse

```
LoginResponse:
  type: object

  properties:

    access_token:
      type: string

    refresh_token:
      type: string

    expires_in:
      type: integer
```

USER SCHEMA

```
User:
  type: object

  properties:

    id:
      type: string

    name:
      type: string

    email:
      type: string

    avatar_url:
      type: string

    status:
      type: string
```

ORGANIZATION SCHEMA

```
Organization:
  type: object

  properties:

    id:
      type: string

    name:
      type: string

    slug:
```

```
    type: string

    industry:
      type: string

    timezone:
      type: string
```

SOCIAL ACCOUNT SCHEMA

```
SocialAccount:
  type: object

  properties:

    id:
      type: string

    platform:
      type: string

    account_name:
      type: string

    status:
      type: string
```

POST SCHEMA

```
Post:
  type: object

  properties:

    id:
      type: string

    title:
      type: string

    caption:
      type: string

    image_url:
      type: string

    status:
      type: string

    created_at:
```

type: string

CREATE POST REQUEST

CreatePostRequest:

type: object

required:

- title
- caption

properties:

title:

type: string

caption:

type: string

image_url:

type: string

platforms:

type: array

items:

type: string

SCHEDULE REQUEST

SchedulePostRequest:

type: object

required:

- post_id
- schedule_time

properties:

post_id:

type: string

schedule_time:

type: string

timezone:

type: string

AI GENERATION REQUEST

Caption

```
GenerateCaptionRequest:  
  type: object  
  
  required:  
    - keyword  
  
  properties:  
  
    keyword:  
      type: string  
  
    platform:  
      type: string  
  
    tone:  
      type: string  
  
    language:  
      type: string
```

Caption Response

```
GenerateCaptionResponse:  
  type: object  
  
  properties:  
  
    caption:  
      type: string  
  
    score:  
      type: integer  
  
    tokens_used:  
      type: integer
```

HASHTAG RESPONSE

```
HashtagResponse:  
  type: object  
  
  properties:  
  
    hashtags:
```

```
type: array

items:
  type: string
```

SEO RESPONSE

```
SeoResponse:
  type: object

properties:

  title:
    type: string

  meta_description:
    type: string

  keywords:
    type: array
```

ANALYTICS SCHEMA

```
Analytics:
  type: object

properties:

  reach:
    type: integer

  impressions:
    type: integer

  clicks:
    type: integer

  engagement:
    type: number
```

BILLING SCHEMA

Plan

```
Plan:
  type: object
```

```
properties:

  id:
    type: string

  name:
    type: string

  price:
    type: number

  max_accounts:
    type: integer

  max_posts:
    type: integer
```

PAYMENT SCHEMA

```
Payment:
  type: object

  properties:

    id:
      type: string

    amount:
      type: number

    status:
      type: string

    payment_gateway:
      type: string
```

WEBHOOK CONTRACTS

Stripe Webhook

POST /webhooks/stripe

Headers:

Stripe-Signature

Payload:

```
{  
  "event": "payment_success",  
  "subscription_id": "uuid"  
}
```

Razorpay Webhook

POST /webhooks/razorpay

Headers:

X-Razorpay-Signature

Facebook Webhook

POST /webhooks/facebook

Verify:

hub.challenge
hub.verify_token

INTERNAL WORKER CONTRACTS

Not exposed publicly.

Publish Worker

POST /internal/jobs/publish

Request

```
{  
  "post_id": "uuid"  
}
```

Analytics Sync Worker

POST /internal/jobs/analytics-sync

Request

```
{  
  "platform": "facebook"  
}
```

Refresh Token Worker

POST /internal/jobs/token-refresh

Request

```
{  
  "account_id": "uuid"  
}
```

RATE LIMIT CONTRACT

Public APIs

20 requests/minute

Authenticated APIs

100 requests/minute

Admin APIs

500 requests/minute

Headers:

X-RateLimit-Limit

X-RateLimit-Remaining

X-RateLimit-Reset

AUDIT CONTRACT

Every critical action logs:

user_id

action
module
old_value
new_value
ip_address
timestamp

Events:

Login
Password Change
Billing Change
Role Change
Post Publish
Subscription Update

FILE UPLOAD CONTRACT

Content Type:

multipart/form-data

Supported:

jpg
jpeg
png
webp
mp4
pdf

Max File Size:

50 MB

Response:

```
{  
  "success": true,  
  "url": "https://cdn.smartautopost.com/file.png"  
}
```

API VERSIONING POLICY

Current:

/api/v1

Future:

/api/v2

/api/v3

Rules:

Breaking changes require new version.

Old version supported minimum 12 months.

SWAGGER TAGS

Auth

Organizations

Teams

BrandKit

SocialAccounts

Keywords

AI

Media

Posts

Scheduler

Publishing

Analytics

Competitors

Festivals

Notifications

Billing

Clients

Admin

Webhooks

Internal

RELEASE DEFINITION

Swagger documentation is considered complete when:

100% endpoints documented
100% request schemas defined
100% response schemas defined
All error codes documented
Webhook contracts documented
Authentication documented
Rate limits documented

SmartAuto Post

Part-13 AI Prompt Library (Production Prompt System)

Version: 1.0

Purpose:

Ye document actual production-grade prompts define karega jo AI Engine use karega.

Is document ki quality directly affect karegi:

Caption Quality
Hashtag Quality
SEO Quality
Engagement Rate
User Retention
AI Cost

AI PROMPT SYSTEM ARCHITECTURE

User Input
↓
Prompt Builder
↓
Prompt Template
↓
LLM
↓
Output Formatter
↓
Quality Checker
↓
Response

MASTER SYSTEM PROMPT

Har AI request me prepend hoga.

You are an expert Social Media Strategist, SEO Specialist and Content Marketing Consultant.

Generate content that is:

- Human sounding
- Engaging
- Platform optimized
- Non repetitive
- Action oriented
- Conversion focused

Avoid:

- Generic content
- Overused marketing buzzwords
- Spammy hashtags
- Robotic writing

Always optimize content for maximum engagement and readability.

PROMPT 1

CAPTION GENERATOR

Input Variables

```
{  
  "keyword":"CRM Software",  
  "platform":"linkedin",  
  "tone":"professional",  
}
```

```
"industry":"SaaS",
"audience":"Business Owners",
"language":"English"
}
```

LinkedIn Caption Prompt

Generate a high-performing LinkedIn post.

Keyword:
{{keyword}}

Industry:
{{industry}}

Audience:
{{audience}}

Tone:
{{tone}}

Requirements:

1. Start with a strong hook.
2. Explain a business problem.
3. Present a practical solution.
4. Keep readability high.
5. Add a strong CTA.
6. Maximum 180 words.
7. No emojis.
8. Professional style.

Return only the caption.

Instagram Caption Prompt

Generate an engaging Instagram caption.

Keyword:
{{keyword}}

Requirements:

1. Start with an attention-grabbing line.
2. Use emojis naturally.
3. Keep conversational tone.
4. Add CTA.
5. Add line breaks.
6. Maximum 150 words.

Return only caption.

Facebook Caption Prompt

Generate a storytelling Facebook post.

Keyword:
{{keyword}}

Requirements:

Problem
Story
Solution
CTA

Maximum 250 words.

PROMPT 2

HASHTAG GENERATOR

Input:

```
{  
  "keyword":"CRM Software",  
  "platform":"instagram"  
}
```

Prompt:

Generate 30 optimized hashtags.

Keyword:
{{keyword}}

Requirements:

10 primary hashtags
10 niche hashtags
10 trending hashtags

Rules:

No duplicates
No banned hashtags
Maximum 25 characters each

Return JSON:

```
{
  "primary": [],
  "niche": [],
  "trending": []
}
```

PROMPT 3

CTA GENERATOR

Generate 10 conversion-focused call-to-actions.

Keyword:
{{keyword}}

Industry:
{{industry}}

Audience:
{{audience}}

Requirements:

Short
Action-oriented
High-converting

Return JSON array.

PROMPT 4

SEO CONTENT GENERATOR

Generate SEO content.

Keyword:
{{keyword}}

Requirements:

SEO Title
Meta Description
SEO Keywords
Slug
Excerpt

Rules:

Meta Description:
150-160 chars

Title:
Maximum 60 chars

Return JSON.

PROMPT 5

IMAGE PROMPT GENERATOR

Purpose:

AI Image Creation.

Generate an image generation prompt.

Keyword:
{{keyword}}

Industry:
{{industry}}

Style:
Modern SaaS

Requirements:

Highly detailed
Professional
Marketing focused
Photorealistic

Return image prompt only.

PROMPT 6

CAROUSEL GENERATOR

Generate a LinkedIn carousel.

Keyword:
{{keyword}}

Create:

Slide 1 Hook

Slide 2 Problem

Slide 3 Insight

Slide 4 Solution

Slide 5 Example

Slide 6 CTA

Return JSON.

PROMPT 7

VIDEO IDEA GENERATOR

Generate 20 short-form video ideas.

Keyword:

{{keyword}}

Platform:

Instagram Reels

Requirements:

Hook

Concept

CTA

Return JSON.

PROMPT 8

30-DAY CONTENT CALENDAR

Create a 30-day social media content calendar.

Industry:

{{industry}}

Audience:

{{audience}}

Posting Frequency:
Daily

Generate:

Day
Topic
Content Type
Caption Idea
CTA

Return JSON array.

PROMPT 9

FESTIVAL POST GENERATOR

Generate a festival social media post.

Festival:
{{festival}}

Industry:
{{industry}}

Requirements:

Warm
Brand-friendly
Professional

Generate:

Caption
Hashtags
CTA
Image Prompt

Return JSON.

PROMPT 10

COMPETITOR ANALYSIS PROMPT

Input:

```
{  
  "competitor_data":"..."  
}
```

Prompt:

Analyze competitor social media data.

Provide:

Posting Frequency
Best Content Type
Best Performing Topics
Hashtag Strategy
Engagement Pattern

Give actionable recommendations.

Return structured JSON.

PROMPT 11

CONTENT IMPROVER

Use after first AI output.

Rewrite and improve the following content.

Requirements:

Increase engagement
Improve readability
Improve conversion potential
Remove repetitive phrases

Content:
{{content}}

Return improved version only.

PROMPT 12

CONTENT SCORING ENGINE

Analyze content quality.

Score:

Engagement Score
SEO Score
Readability Score
Conversion Score

Content:
{{content}}

Return JSON:

```
{  
  "engagement":0,  
  "seo":0,  
  "readability":0,  
  "conversion":0,  
  "recommendations":[]  
}
```

PROMPT 13

TREND DETECTOR

Analyze these trending keywords.

Data:
{{trend_data}}

Return:

Top Trends
Content Opportunities
Risky Trends
Suggested Topics

Return JSON.

PROMPT 14

BEST POSTING TIME ENGINE

Analyze post performance data.

Metrics:
{{analytics}}

Return:

Best Day
Best Time
Worst Time
Recommendation

Return JSON.

PROMPT 15

AGENCY CONTENT GENERATOR

Special prompt for agencies.

Generate content for client.

Client Industry:
{{industry}}

Client Goal:
{{goal}}

Target Audience:
{{audience}}

Generate:

Caption
Hashtags
CTA
Image Idea

Return JSON.

AI QUALITY VALIDATION LAYER

AI output publish hone se pehle validation:

Check 1

Minimum Content Length

Check 2

No Duplicate Content

Check 3

No Spam Words

Check 4

No Platform Violations

Check 5

Readability > 70

PROMPT VERSIONING TABLE

prompt_templates

id
name
version
prompt_text
status
created_at

Example:

caption_linkedin_v1

caption_linkedin_v2

caption_linkedin_v3

A/B TESTING STRATEGY

Store:

Prompt Version

Engagement

Clicks

Shares

Conversions

Find:

Best Performing Prompt

AI COST OPTIMIZATION

Instead of:

Generate Full Content

Use:

Template
+
AI Enhancement

Example:

Hook Template
+
AI Personalization

Result:

70-80% Lower AI Cost

PROMPT CACHE KEYS

caption:{keyword}:{platform}

hashtags:{keyword}

seo:{keyword}

calendar:{industry}

TTL:

24 Hours

FUTURE AI AGENT PROMPTS

Agents:

Content Strategist Agent

SEO Agent

Publisher Agent

Analytics Agent

Competitor Agent

Each agent gets its own system prompt and task-specific prompts.
